

BEDFORD 5 WNW, IN, USA

WMO: 724310

Lat: 38.8882N

Lon: 86.5707W

Elev: 232

StdP: 98.57

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 63898

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.3	-19.4	0.7	-14.2	-16.9	0.9	-11.2	5.8	0.8	5.3	-1.2	1.7	N/A	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	32.3	24.0	31.0	23.7	29.8	23.3	25.8	30.1	25.1	29.3	24.3	28.4	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.1	28.7	23.8	19.2	27.8	23.1	18.3	26.8	81.2	30.1	77.7	29.4	74.7	28.4	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.0	4.3	3.8	DB	-18.7	33.7	3.5	2.1	-21.3	35.2	-23.3	36.5	-25.3	37.7	-27.9	39.2	(4)
(5)			WB	-18.8	27.1	3.7	0.8	-21.5	27.6	-23.7	28.1	-25.8	28.5	-28.5	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-1.3	1.4	7.0	13.2	18.1	22.6	23.8	23.2	20.1	13.7	6.7	2.2	(6)
(7)		DBStd	10.08	6.70	6.85	6.21	5.01	4.47	2.92	2.91	2.61	3.86	5.12	5.43	5.81	(7)
(8)		HDD10.0	1164	354	253	134	26	1	0	0	0	0	21	126	251	(8)
(9)		HDD18.3	2705	608	476	354	163	62	4	1	2	25	161	348	501	(9)
(10)		CDD10.0	2116	4	11	41	122	252	377	427	408	304	135	28	8	(10)
(11)		CDD18.3	616	0	1	2	11	54	131	169	151	79	18	1	0	(11)
(12)		CDH23.3	4864	0	1	10	69	387	1028	1437	1172	639	119	2	0	(12)
(13)		CDH26.7	1449	0	0	0	3	66	307	489	358	202	23	0	0	(13)
(14)	Wind	WSAvg	1.7	2.2	2.2	2.3	2.2	1.5	1.3	1.2	1.1	1.2	1.5	1.8	2.0	(14)
(15)	Precipitation	PrecAvg	1133	72	66	99	107	131	106	113	98	74	82	94	85	(15)
(16)		PrecMax	1561	202	153	278	329	281	258	270	214	191	268	219	175	(16)
(17)		PrecMin	817	9	5	25	28	48	9	16	8	2	10	26	14	(17)
(18)		PrecStd	173	44	36	57	58	60	50	58	54	48	53	45	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	20.6	24.9	27.0	30.2	32.6	36.3	33.7	32.7	29.4	22.7	19.0	(19)
(20)			MCWB	15.1	15.4	17.4	17.6	22.8	23.8	23.8	23.7	23.2	20.7	16.0	15.7	(20)
(21)		2%	DB	14.4	17.5	21.9	25.4	28.5	31.0	32.5	31.7	30.9	26.4	20.0	16.3	(21)
(22)			MCWB	11.8	13.5	15.3	17.0	21.7	23.4	24.8	23.4	22.5	20.0	15.1	14.1	(22)
(23)		5%	DB	11.8	14.8	19.1	23.8	27.1	29.9	30.8	30.1	29.2	24.2	17.6	13.5	(23)
(24)			MCWB	9.5	11.9	13.4	16.3	20.7	23.3	24.8	23.2	22.3	18.2	13.1	11.3	(24)
(25)		10%	DB	8.9	12.0	16.6	21.8	25.7	28.5	29.3	28.8	27.1	22.2	15.3	10.7	(25)
(26)			MCWB	6.7	8.9	11.9	15.3	20.0	22.4	23.9	22.7	21.2	17.3	11.1	8.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	16.6	18.1	20.1	23.8	26.0	27.1	26.6	24.9	22.9	18.2	16.4	(27)
(28)			MCD B	16.7	20.3	24.3	24.9	28.3	30.3	31.5	31.6	30.1	27.8	21.1	18.4	(28)
(29)		2%	WB	12.2	14.7	16.3	18.6	22.7	25.1	26.2	25.3	23.9	21.3	16.0	14.2	(29)
(30)			MCD B	13.6	16.8	20.7	23.2	27.2	29.6	30.4	29.8	29.0	24.4	19.1	15.8	(30)
(31)		5%	WB	10.0	12.4	14.6	17.4	21.8	24.1	25.4	24.5	23.1	19.7	13.7	11.7	(31)
(32)			MCD B	11.3	14.6	18.2	21.2	26.0	28.4	29.4	28.4	27.7	22.8	16.7	13.4	(32)
(33)		10%	WB	6.4	8.9	12.8	16.4	21.0	23.3	24.7	23.7	22.2	17.8	11.4	8.7	(33)
(34)			MCD B	8.9	11.3	15.7	20.0	24.9	27.3	28.5	27.1	25.8	20.8	14.7	10.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	9.5	10.5	9.6	9.7	9.3	10.0	10.4	10.3	9.2	7.5	(35)
(36)			MCDBR	10.9	11.8	11.9	11.9	10.8	11.0	10.8	11.3	12.2	11.9	11.6	10.6	(36)
(37)			MCWBR	9.3	9.7	7.5	6.9	5.5	5.5	4.6	4.8	5.2	6.0	7.6	8.7	(37)
(38)		5% WB	MCD BR	10.5	11.5	10.7	10.3	9.6	9.6	9.3	9.1	10.2	9.7	10.1	9.3	(38)
(39)			MCWBR	10.0	10.3	7.4	6.9	5.1	5.3	4.8	4.7	5.0	5.6	8.0	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.322	0.351	0.400	0.432	0.444	0.461	0.460	0.418	0.355	0.323	0.301	(40)	
(41)		taud	2.467	2.394	2.386	2.253	2.208	2.209	2.161	2.154	2.255	2.442	2.477	2.487	(41)	
(42)		Ebn at Noon	871	901	906	876	852	839	820	811	824	852	842	850	(42)	
(43)		Edh at Noon	80	98	110	133	143	143	148	146	124	93	79	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.85	2.61	3.67	4.82	5.41	6.23	5.97	5.53	4.56	3.29	2.22	1.56	(44)	
(45)		RadStd	0.20	0.31	0.34	0.33	0.50	0.39	0.37	0.33	0.38	0.34	0.24	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	N/A

Nomenclature: See separate page